Sentiment Analysis Of Comments Received Through E-Consultation Module

ABSTRACT

The rapid adoption of digital governance platforms has led to a significant increase in public participation through online consultation systems. The e-consultation module enables citizens and stakeholders to submit comments, suggestions, and feedback on policies and services. However, the large volume of unstructured textual data generated through this module makes manual analysis inefficient and time-consuming. To address this challenge, this project focuses on performing **sentiment analysis of comments received through the e-consultation module**.

The proposed system automatically analyzes user-submitted comments and classifies them into sentiment categories such as **positive, negative, and neutral** using Natural Language Processing (NLP) techniques. Text preprocessing steps including tokenization, stop-word removal, and normalization are applied to improve data quality. Machine learning and deep learning models are employed to extract meaningful patterns and determine the underlying sentiment of each comment. The analyzed results provide valuable insights into public opinion, helping decision-makers understand trends, concerns, and overall sentiment efficiently.

This system enhances transparency and data-driven decision-making by transforming raw textual feedback into actionable insights. It reduces manual effort, improves response time, and supports policymakers in evaluating public sentiment more accurately. The solution is scalable and can be extended to support multilingual data and real-time sentiment monitoring in future implementations.

INTRODUCTION

eConsultation module is an online platform wherein proposed amendments/draft legislations are posted on MCA's website for external users to submit their comments and suggestions pertaining to the same through the MCA21 portal. The comments are captured in a structured format for due consideration with respect to amending the draft legislation, based on the suggestions or observations received.

The draft document soliciting comments is made available for a specified period, during which any stakeholder may submit their observations either on the overall amendment or on specific provisions of the draft legislation. In instances where a substantial volume of comments is received on draft legislation, there exists a risk of certain observations being inadvertently overlooked or inadequately analysed. In order to review each individual submission, leveraging AI-assisted tools will help ensure that all remarks are duly considered and systematically analysed. Requirement is the development of an AI model aimed at predicting the sentiments of the suggestions provided by stakeholders in the eConsultation module. It should also generate a visual representation in the form of a word cloud, highlighting the keywords utilised by the stakeholders within their suggestions.

PROBLEM STATEMENT

With the increasing adoption of digital platforms in governance and public service delivery, **e-consultation modules** have become a vital medium for collecting feedback, opinions, and suggestions from citizens and stakeholders. These platforms allow users to submit comments on policies, schemes, regulations, and public services, resulting in the generation of a large volume of **unstructured textual data**. While this data contains valuable insights into public opinion and sentiment, it is often underutilized due to the absence of efficient analytical mechanisms.

Currently, the analysis of comments received through e-consultation modules is largely **manual or semi-automated**, requiring significant human effort, time, and resources. Manual review becomes impractical as the number of comments grows, leading to delays in decision-making and an increased risk of bias or inconsistency in interpretation. Additionally, traditional keyword-based approaches fail to capture the contextual meaning, emotional tone, and sentiment expressed in natural language, especially when comments are lengthy, informal, or ambiguous.

Another major challenge lies in handling **noise in textual data**, such as spelling errors, slang, abbreviations, emojis, and mixed-language usage, which reduces the accuracy of conventional text-processing techniques. The lack of real-time sentiment monitoring further limits the ability of authorities to quickly identify emerging trends, public concerns, or critical feedback that may require immediate attention.

Due to these limitations, policymakers and administrators struggle to extract actionable insights from citizen feedback, which impacts the effectiveness of public engagement and data-driven governance. There is a clear need for an **automated, scalable, and intelligent system** that can efficiently analyze large volumes of comments, accurately determine the sentiment expressed, and present the results in a meaningful and interpretable manner.

Therefore, the problem addressed in this project is the **absence of an efficient sentiment analysis mechanism for automatically classifying and interpreting comments received through e-consultation modules**. The proposed solution aims to leverage Natural Language Processing and machine learning techniques to analyze user comments, categorize them into sentiment classes such as positive, negative, and neutral, and assist decision-makers in understanding public opinion quickly and accurately.

LITERATURE SURVEY

Various sentiment analysis techniques have been explored in existing research, ranging from rule-based approaches to advanced deep learning models. **VADER**, a lexicon-based method, is widely used for its speed and ease of implementation, as it does not require labeled training data. However, its performance degrades when handling slang, informal language, or code-mixed text, which is common in e-consultation comments.

Traditional machine learning models such as **Naive Bayes** utilize probabilistic principles and offer simplicity and low computational cost. Despite these advantages, Naive Bayes relies on the assumption of feature independence, which often does not hold true in natural language data, leading to reduced classification accuracy.

Logistic Regression has been extensively adopted in sentiment classification tasks due to its discriminative nature and ability to efficiently handle high-dimensional textual features such as TF-IDF vectors. It provides robust performance, faster training time, and better generalization compared to basic probabilistic models. Moreover, Logistic Regression offers interpretability of results, making it suitable for real-world applications where transparency is essential. These characteristics make it a practical and balanced choice for analyzing large-scale e-consultation data.

Recent advancements in deep learning, particularly **BERT-based transformer models**, have achieved state-of-the-art accuracy by capturing contextual and semantic relationships in text. However, their high computational requirements, longer training time, and dependency on extensive resources limit their feasibility for lightweight and scalable deployment in many practical systems.

Based on the comparative analysis, **Logistic Regression emerges as an effective trade-off between performance, efficiency, and interpretability**, making it well-suited for sentiment analysis of comments received through e-consultation modules.

PROPOSED MODEL

The proposed system is designed to perform automated sentiment analysis on public opinions related to governance and policy discussions. Initially, the project intended to analyze comments collected directly from the MCA21 e-consultation portal. However, due to the inaccessibility and unavailability of public comment data from the MCA21 portal, it was not feasible to retrieve sufficient real-time feedback for analysis. To address this limitation, an alternative and widely used public discussion platform was selected.

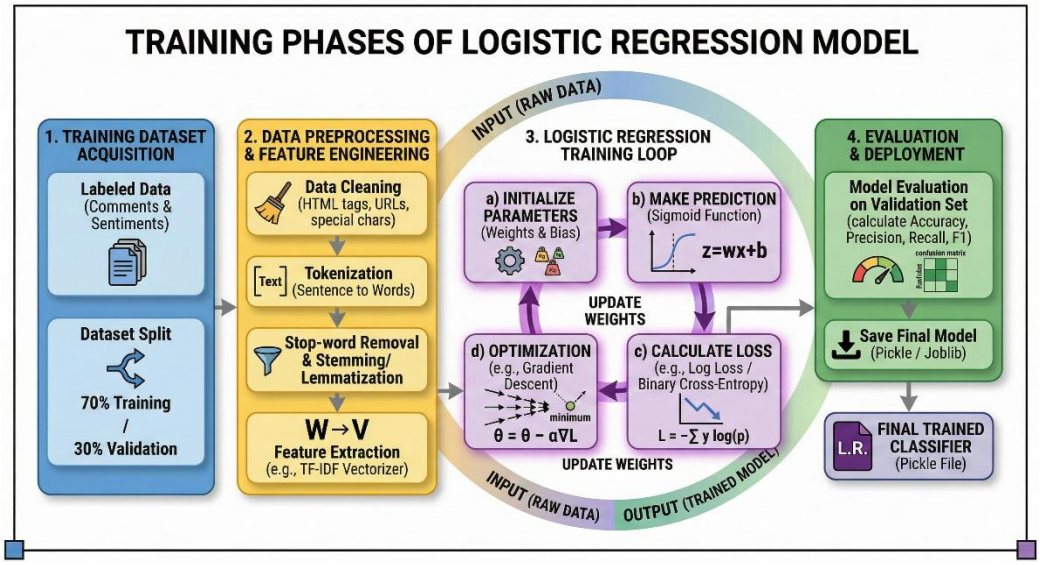
YouTube was chosen as the data source, as it serves as a major platform where a large number of citizens actively express their opinions, feedback, and reactions through comments on videos related to government policies, public announcements, and social issues. The comments posted on YouTube reflect diverse public sentiment and closely resemble the feedback typically collected through formal e-consultation modules. Therefore, YouTube comments are considered a suitable and realistic substitute for analyzing public opinion in this context.

The proposed system collects user comments from selected YouTube videos . These comments are inherently unstructured and may contain noise such as slang, abbreviations, emojis, and grammatical inconsistencies. To handle this, a robust text preprocessing pipeline is implemented, which includes cleaning, tokenization, stop-word removal, normalization, and lemmatization to improve data quality.

After preprocessing, the cleaned textual data is converted into numerical feature vectors using TF-IDF representation. These feature vectors are then used to train a Logistic Regression–based sentiment classification model, which categorizes comments into positive, negative, and neutral sentiment classes. Logistic Regression is chosen due to its efficiency, scalability, and ability to handle high-dimensional text data while providing interpretable results.

The system automatically analyzes large volumes of comments and generates sentiment-based insights that reflect public opinion trends. The final output enables stakeholders to understand overall sentiment patterns, identify key concerns, and evaluate public response effectively. By

using YouTube comments as an alternative data source, the proposed system successfully demonstrates a practical and scalable approach for sentiment analysis in scenarios where direct e-consultation data is unavailable.



In addition to sentiment classification, the proposed system incorporates an **opinion mining module** that focuses on identifying key opinions, recurring viewpoints, and dominant themes expressed by the public. This module analyzes frequently occurring terms, sentiment-bearing phrases, and contextual patterns to extract meaningful opinions from large sets of comments. Opinion mining enables the system to go beyond simple polarity detection and capture what specific aspects or issues users are expressing opinions about.

The system also generates **word clouds** to visually represent frequently occurring keywords and opinion-related terms in the collected comments. These word clouds provide an intuitive overview of dominant topics and concerns discussed by the public.

The final output of the proposed system includes sentiment classification results, extracted opinions, and visual word cloud representations. Together, these components provide a structured and interpretable understanding of public opinion, reduce manual analysis effort, and support data-driven decision-making. By leveraging YouTube comments as an alternative data source, the proposed system demonstrates a practical and scalable approach to sentiment analysis and opinion mining in scenarios where direct e-consultation data is unavailable.

SYSTEM ARCHITECTURE

The system architecture illustrates the overall structure and interaction between different modules involved in the sentiment analysis and opinion mining process. The proposed system follows a modular and layered architecture to ensure scalability, maintainability, and efficient processing of large volumes of textual data.

The architecture begins with **Data Source.**Kaggle contains a dataset with 17k data points that better suites to train the model for sentiment prediction.Along with this mock data is added to train the model on code mixed languages.Since the MCA21 e-consultation portal is

inaccessible, YouTube serves as a reliable alternative platform that provides diverse and real-time public opinions.

The **Data Collection Module** is responsible for retrieving comments using platform APIs. Youtube provides its comments with the help of **YOUTUBE V3 API** . The collected data is stored in a structured format for processing.

Next, the **Text Preprocessing Module** cleans and normalizes the raw comments. This module removes noise such as punctuation, URLs, emojis, and stop words, and performs tokenization to prepare the text for analysis.

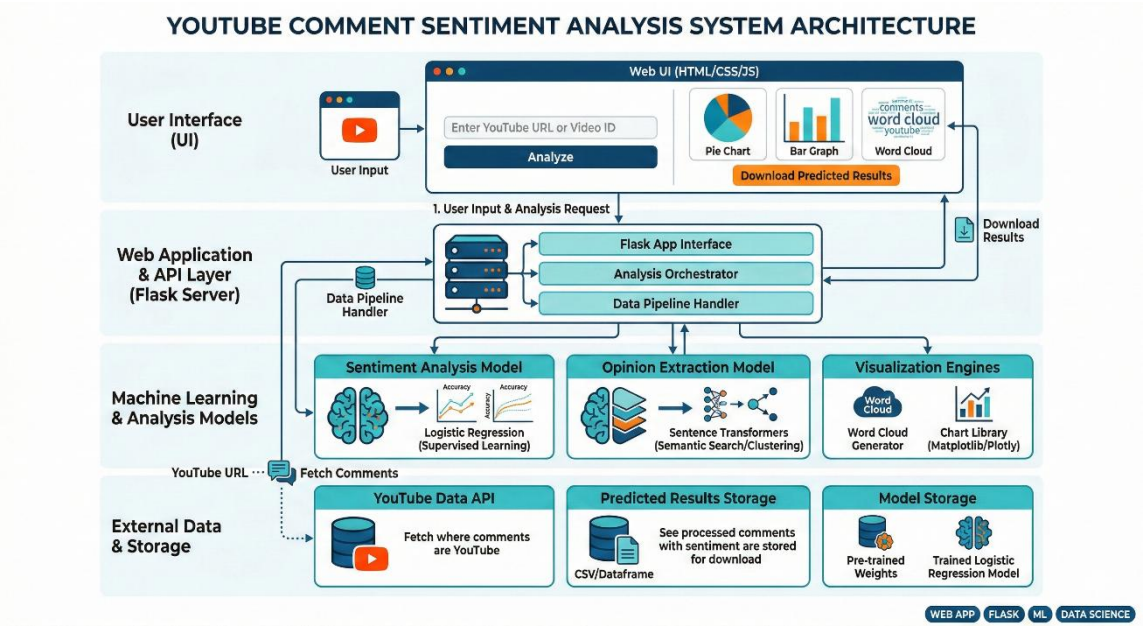
The processed text is then passed to the **Feature Extraction Module**, where meaningful numerical representations are generated using TF-IDF vectorization. These feature vectors capture the importance of words and reduce dimensionality.

The **Sentiment Classification Module** employs a Logistic Regression model to classify each comment into positive, negative, or neutral sentiment categories. This module provides accurate and interpretable sentiment predictions with low computational overhead.

In parallel, the **Opinion Mining Module** analyzes sentiment-bearing phrases and frequently occurring terms to extract dominant opinions and themes expressed by users.

The **Visualization Module** generates word clouds and summary statistics to visually represent public opinion and sentiment trends.

Finally, the **Output Layer** presents sentiment results, extracted opinions, and visual insights to users or administrators for informed decision-making.



METHODOLOGY

The methodology of the proposed system follows a systematic sequence of steps to ensure accurate sentiment analysis and opinion mining.

1. **Data Collection**
   User enters a video URL or video_id. Youtube V3 API helps to extract the raw comments based on the video_id. The collected data is stored in a structured format to process further.

2. **Data Preprocessing**
   Raw comments are cleaned by removing noise including special characters, URLs, emojis, and stop words. The text is converted to lowercase, tokenized to improve consistency and reduce redundancy.

3. **Feature Extraction**
   The preprocessed text is transformed into numerical feature vectors using TF-IDF. This method highlights important words while minimizing the influence of frequently occurring but less informative terms.

4. **Sentiment Classification**
   A Logistic Regression model is trained on labeled data and used to classify comments into positive, negative, or neutral sentiment categories. The model's efficiency and interpretability make it suitable for large-scale analysis.

5. **Opinion Mining**
   The system identifies key opinion phrases, frequently discussed topics, and sentiment-specific patterns to extract meaningful opinions from user comments.

6. **Word Cloud Generation**
   Word clouds are generated based on word frequency to visually highlight dominant terms and themes. Separate word clouds may be created for each sentiment category.

7. **Result Analysis and Visualization**
   Sentiment distribution, extracted opinions, and visual representations are analyzed to understand public opinion trends and overall sentiment.

TECHNOLOGIES USED

Category	Technology / Tool	Purpose
Programming Language	Python	Core language for sentiment analysis and opinion mining
Data Source	YouTube Data API	Collection of public comments
Text Processing	NLTK	Text cleaning, tokenization, and normalization

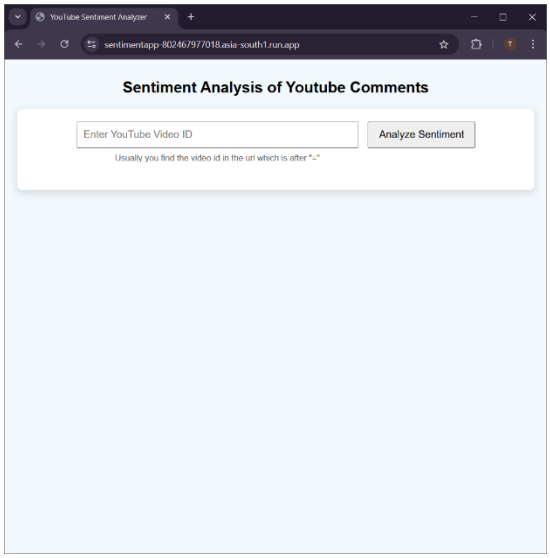
Category	Technology / Tool	Purpose
Feature Extraction	TF-IDF (scikit-learn)	Conversion of text into numerical features
Machine Learning Model	Logistic Regression	Sentiment classification (positive, negative, neutral)
Visualization	WordCloud, Matplotlib	Opinion visualization and sentiment trends

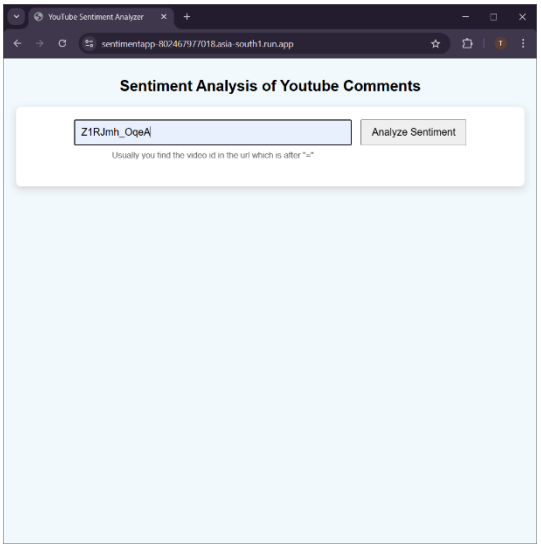
RESULTS AND CONCLUSION

The proposed system was successfully implemented to analyze public opinions using comments collected from YouTube as an alternative to the MCA21 e-consultation portal. The system efficiently processed a large volume of unstructured textual data and accurately classified comments into **positive, negative, and neutral** sentiment categories using a Logistic Regression model.

The sentiment analysis results provided a clear distribution of public opinion, enabling the identification of dominant sentiment trends related to the selected topics. The opinion mining module extracted frequently expressed viewpoints and key terms, offering deeper insight into the aspects that received strong public attention. Additionally, **word cloud visualizations** effectively highlighted commonly discussed keywords, making it easier to understand major concerns and positive responses at a glance.

The results demonstrate that the proposed approach significantly reduces manual effort required for analyzing feedback and delivers consistent and interpretable sentiment insights. The system performed efficiently with low computational overhead while maintaining reliable classification accuracy, making it suitable for large-scale opinion analysis.





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...              precision    recall  f1-score   support

           0       0.76      0.97      0.85        38
           1       0.67      0.77      0.72        43
           2       0.92      0.69      0.79        67

    accuracy                           0.78       148
   macro avg       0.78      0.81      0.78       148
weighted avg       0.81      0.78      0.78       148
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Deployed URL : https://sentimentapp-802467977018.asia-south1.run.app/

CONCLUSION

This project successfully demonstrates an automated approach for **sentiment analysis and opinion mining** of public comments in scenarios where direct access to formal e-consultation platforms such as the MCA21 portal is unavailable. By leveraging YouTube comments as an alternative data source, the system effectively captures diverse public opinions and provides meaningful insights into sentiment trends.

The use of Natural Language Processing techniques combined with a **Logistic Regression–based classification model** ensures efficient, scalable, and interpretable sentiment analysis. The inclusion of opinion mining and word cloud visualization further enhances the understanding of public viewpoints by highlighting key themes and frequently discussed topics.

Overall, the proposed system offers a practical and cost-effective solution for analyzing large volumes of public feedback. It supports data-driven decision-making, improves transparency, and can serve as a foundation for future enhancements such as multilingual support, real-time sentiment monitoring, and integration with official e-consultation platforms.

FUTURE ENHANCEMENTS

Although the proposed system effectively performs sentiment analysis and opinion mining on public comments, several enhancements can be implemented to further improve its functionality and applicability.

1. **Multilingual Sentiment Analysis :** The system can be extended to support comments in multiple languages and code-mixed text, enabling analysis of opinions from a more diverse population.
2. **Aspect-Based Opinion Mining :** Instead of analyzing overall sentiment, future work can focus on identifying sentiment toward specific aspects or topics within comments, providing more granular insights.
3. **Advanced Deep Learning Models :** The current Logistic Regression model can be replaced or enhanced with advanced deep learning models such as LSTM or transformer-based models to improve contextual understanding and classification accuracy.
4. **Real-Time Sentiment Monitoring :** The system can be upgraded to analyze comments in real time, allowing authorities to track sentiment trends and public reactions as they evolve.
5. **Integration with Official E-Consultation Platforms :** In the future, the system can be integrated directly with official platforms such as the MCA21 e-consultation portal when data access becomes available.
6. **Interactive Dashboard and Analytics :** A web-based dashboard can be developed to visualize sentiment trends, opinion patterns, and word clouds dynamically, improving usability for decision-makers.